

PRADEEP KUMAR BALASUBRAMANIAN

Embedded Software Engineer — Automotive Telematics, RTOS & Bootloaders

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PROFESSIONAL SUMMARY

Embedded Software Engineer with 4+ years of hands-on experience developing mission-critical firmware in C, C++, and Python across automotive and industrial systems. Strong background in RTOS-based deterministic design, custom secondary bootloaders, flash memory management, and automotive diagnostic protocols (CAN 2.0B, J1939, UDS ISO 15765-4, XCP, K-Line). Proven production experience deploying telematics units, managing FOTA campaigns, and engineering rugged industrial IoT controllers.

TECHNICAL SKILLS

CORE LANG: Embedded C (MISRA-C), C++ (C++14/17), Python 3.x, Bash / Shell, Assembly (ARM)

MCU PLATFORM: ARM Cortex-M (STM32), Renesas RA Series (M33), Renesas RH850, ESP32 / ESP8266, Microchip PIC, Quectel EC200 (LTE Cat 1), Raspberry Pi, Arduino

RTOS SYSTEM: FreeRTOS, QL_RTOS (QuecOpen), Custom Secondary Bootloaders, Deterministic Task Scheduling, Queues / Mutexes / Semaphores, Flash Partitioning / Dual-Bank, Low-Power Modes (Sleep/Deep Sleep), ISR Optimization

BUS PROTOCOL: CAN 2.0B, SAE J1939, ISO 15765-4 (UDS / ISO-TP), XCP on CAN, K-Line (ISO 9141), SAE J1708, UART / RS-485 / RS-232, SPI (High-Speed), I2C, MODBUS RTU, MQTT

DEBUG INTEG: Logic Analyzers (Saleae / DSLogic), Digital Storage Oscilloscopes, JTAG / SWD (J-Link / ST-LINK), System-Level Fault Analysis, Flash Memory Wear Leveling, Automated FOTA Validation, EMI / ESD Debugging

TOOLCHAIN IDE: Renesas e2 studio, STM32CubeIDE / CubeMX, Keil µVision, VS Code, Git / GitHub, MATLAB / Simulink, ROS (Robot OS), AutoCAD

WORKFLOW ENV: Linux (Ubuntu / Debian), Windows, Jira / Confluence, Agile / Scrum, Peer Code Review, Hardware Homologation Support

WORK EXPERIENCE

Embedded Systems Engineer

Aug 2024 – Present

Intangles Lab Pvt Ltd

Pune, India

Target Hardware / MCU Platform: Renesas RA Series (ARM Cortex-M33) & Quectel EC200 LTE

- Engineered telematics firmware and custom secondary bootloaders on Renesas RA Series MCUs (Cortex-M33), designing flash partition layouts for dual-bank rollback, image authentication flags, and persistent configuration EEPROM emulation.
- Architected Quectel EC200 (QuecOpen / QL_RTOS) cellular SoC firmware to coordinate asynchronous CAN polling, multi-threaded ECU data pipelines, cloud telemetry streaming, and fail-safe FOTA updates.
- Integrated and validated high-speed automotive bus protocols: CAN 2.0B, SAE J1939, ISO 15765-4 (UDS diagnostics), XCP calibration, K-Line (ISO 9141), and J1708 for commercial vehicle diagnostics.
- Conducted bare-metal and OS-level debugging with logic analyzers and oscilloscopes, mitigating bus-off conditions, buffer overruns, and interrupt latency to ensure 99.98% communication reliability.
- Created a comprehensive Python desktop automation framework to simulate ECU diagnostic responses and automate full-cycle FOTA verification testing.
- Collaborated closely with cross-functional hardware, cloud backend, and validation teams for end-to-end device homologation and production scaling.

Embedded Engineer

Jul 2020 – Aug 2022

Nanotronics Scientific Solutions

Coimbatore, India

Target Hardware / MCU Platform: STM32 (ARM Cortex-M4), ESP32, PIC Microcontrollers

- Developed production-grade embedded firmware for automated fuel dispensing kiosks, fluid metering units, and industrial automation equipment in C/C++.
- Implemented robust multi-drop industrial communication protocols including MODBUS RTU over RS-485, multi-channel UART, SPI sensor interfaces, and MQTT telemetry over Wi-Fi.
- Integrated FreeRTOS kernel on STM32 and ESP32 targets, implementing deterministic task scheduling, inter-task message queues, mutexes, and low-jitter software timers.
- Authored hardware abstraction layer (HAL) drivers for flash memory, graphical LCD controllers, flow pulsers, optical encoders, and thermal receipt printers.
- Resolved critical hardware-software race conditions and EMI-induced system resets during high-voltage switching, increasing field MTBF significantly.

Project Intern — Range-Extended EV

Jul 2019 – Aug 2019

SAEINDIA (in collaboration with General Motors)

Bengaluru, India

Target Hardware / MCU Platform: NXP / Microchip Automotive MCUs

- Led the electronics and embedded control team for a student-built Range-Extended Electric Vehicle (REEV) initiative co-sponsored by General Motors.
- Engineered state-machine control logic for automated ICE range-extender generator start/stop sequencing and high-voltage contactor safety interlocks.
- Synthesized CAN telemetry packets between custom motor controller, battery management system (BMS), and driver instrumentation cluster.
- Awarded "Best Performance" and National Runner-Up accolade among 40+ engineering universities across India.

EDUCATION & RESEARCH PUBLICATION

University of York

Sep 2022 – Dec 2023

Postgraduate Certificate, Intelligent Robotics

York, United Kingdom

Sri Shakthi Institute of Engineering and Technology

Jul 2016 – Nov 2020

B.E., Electronics and Communication Engineering

Coimbatore, India

Peer-Reviewed Publication: "Automotive Electronic System Controlling the Range Extended Electric Vehicle" (2020)

International Journal of Engineering Research & Technology (IJERT), ICEECT 2020, Volume 8, Issue 17.